BOMI KWON

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Research Interests: Diffusion-based generative models, structured & controllable generation, safe generation

EDUCATION

Sookmyung Women's University — Seoul, Republic of Korea

Master of Science, Department of Computer Science — Mar. 2025 – Feb. 2027 (expected)
Data Intelligence Laboratory; Advisor: Prof. Ki Yong Lee

Bachelor of Science, Department of Statistics — Mar. 2020 – Feb. 2025
Bachelor of Business Administration, Department of Business Administration

PUBLICATIONS & PRESENTATIONS

B. Kwon and K. Y. Lee (2026). *Text-Conditioned Scene Graph Sequence Generation via Discrete Diffusion.* Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM '26). Accepted Short Paper. [**Paper**]

B. Kwon and K. Y. Lee (2026). *Text-driven 2D Human Pose Generation via Diffusion in Directional Angle Space.* The Transactions of the Korea Information Processing Society, 15(7), 610–619. [**Paper**]

B. Kwon, S. Park, Y. Park, and K. Y. Lee (2026). *Text-Conditioned Diffusion for Humanoid Robot Pose Generation with Joint-Limit Embedding and Self-Collision Guidance.* The 7th Korea Artificial Intelligence Conference (Korea AI 2026). [**Best Paper Award**] [**Paper**]

B. Kwon and K. Y. Lee (2026). *Diffusion-based Skeleton Pose Editing.* 2026 Spring Conference of the Korean Institute of Smart Media and the Korean Society of Digital Industry. [**Paper**]

B. Kwon and K. Y. Lee (2025). *A Graph Generation Model for Text-based Graph Editing.* Korea Software Congress 2025 (KSC 2025). [**Paper**]

B. Kwon and K. Y. Lee (2025). *Development of a Large-Scale Graph Retrieval System based on LLM and Graph Database Providing a Natural Language Interface.* Korea Computer Congress 2025 (KCC 2025). [**Paper**]

HONORS & FUNDING

Program Council Chair's Award – 2026 ICT Talent Development Best-Practice Essay Competition — 2026
Institute for Information & Communications Technology Planning & Evaluation (IITP)

- Selected through external expert review of academic and project achievements in a government-supported ICT talent development program; invited to present at the 2026 outcomes forum.

Best Paper Award – Korea AI 2026 — Oct. 2026
The Korean Institute of Communications and Information Sciences (KICS)

WISET Competitive Research Grant — Apr. 2026 – Oct. 2026
Korea Foundation for Women in Science, Engineering and Technology (WISET) [**Official News**]

- Selected as one of 50 research teams nationwide; awarded KRW 7,000,000 (approx. USD 5,000) in competitive research funding as Principal Investigator.

Sookmyung Leadership Activity Scholarship — Jan. 2023

RESEARCH EXPERIENCE

2026 STEM Research Team Project
Principal Investigator — Apr. 2026 – Oct. 2026

- Lead a team of two undergraduate researchers developing synthetic human-pose generation methods for industrial safety.
- Manage research execution, budget administration, and project reporting for the funded research project.

PROJECTS

AISUM Industry–Academic Collaboration — Product Matching and Retrieval System — 2025 – Present

- Built large-scale visual and semantic product retrieval pipelines using embedding models, Qdrant, and multi-stage ranking for millions of product records.
- Integrated retrieval modules into an MCP-based interface and developed a LangGraph agent workflow for multi-stage product search and recommendation.

Chronology-Aware Scene Graph Sequence Generation from Text 2025 – Present

- Developed a slot-time discrete diffusion framework that jointly generates objects and relations over time while resolving temporal order and overlap in multi-event descriptions.

Cross-Robot Text-Conditioned Humanoid Pose Generation 2026 – Present

- Developed a diffusion-based pose generation framework with joint-limit and self-collision constraints, and extended it across eight humanoid robots to investigate generalization to unseen robot morphologies.

Text-to-Pose Generation in Directional Angle Space 2025 – 2026

- Implemented a BGE-M3-conditioned diffusion pipeline for generating 2D human poses using a directional-angle representation.

Large-Scale Graph Retrieval System with LLM and Neo4j 2025

- Built an LLM-based natural-language interface that translates user requests into Cypher queries for retrieval over large-scale Neo4j graph data.

ENTREPRENEURIAL EXPERIENCE

fitapat (ZOOC)
Founding Team Member / Product & Marketing Lead Jul. 2023 – Aug. 2024

- Co-founded and led product development for an AI-based pet customization service that generated personalized characters from user-provided photos and connected them to customized merchandise.
- Led product and go-to-market execution through launch; the platform and physical products were showcased at CES 2024.

LEADERSHIP & ACTIVITIES

SOPT
Vice President, 32nd Cohort Jan. 2023 – Jul. 2023
Product Manager, 31st Cohort Jul. 2022 – Jan. 2023

- Led a nationwide university IT and startup community; previously served as Product Manager for ZOOC, a digital service for pet owners.

Sookmyung Admissions Ambassador "Polaris" — Vice President Mar. 2021 – Feb. 2023
Statistics Korea (KOSTAT) — Volunteer Statistics Educator May 2021 – Dec. 2021
Amorepacific × Sookmyung A.I. Team Project — Team Project Oct. 2024 – Dec. 2024
SOOK-TAT — Statistics Data Analysis Society, Member Sep. 2022 – Jul. 2023
Department of Statistics Student Council — Member Mar. 2020 – Dec. 2021
KB Campus Star — Student Ambassador May 2022 – Nov. 2022

TECHNICAL SKILLS

Programming: Python, R, SQL
ML Frameworks: PyTorch, Hugging Face Transformers
Retrieval & Databases: Qdrant, PostgreSQL/pgvector, Neo4j
Agentic AI: LangChain, LangGraph, MCP
Simulation: MuJoCo